Numerical Integration

MATH-151: Mathematical Algorithms in Matlab

September 25, 2023



Integration in Practice

- Reminder: The definite integral of a function $f(x)$ between points a and b outputs the area beneath that curve between $x = a$ and $x = b$ and is represented

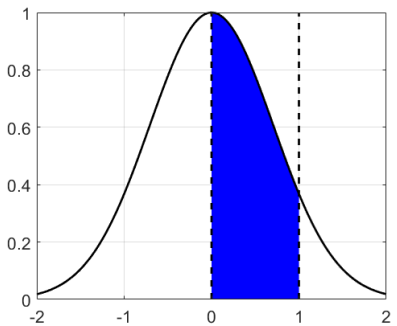
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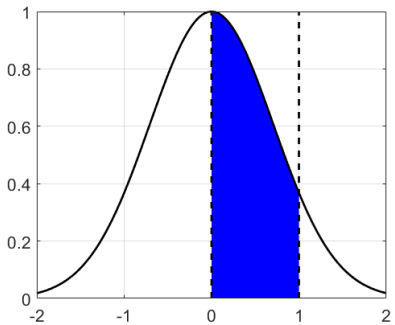
- Integrals effectively "accumulate" the effect of f, for a few examples
  - Integrating velocity over some time tells us how far an object traveled
  - The area under a probability distribution gives us a probability
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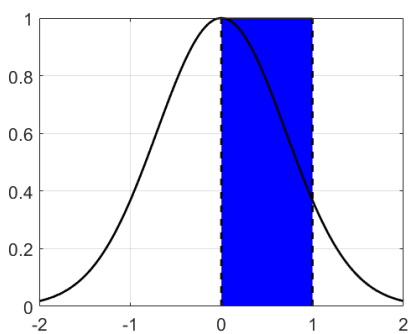
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  - Computing oddly shaped areas and volumes
  - Often used for solutions of differential equations
- There are many different rules for performing these integrals, and many of them are very complicated for a computer to perform, so we find approximations!

Rectangular Approximations

- A very basic way to quickly approximate an integral is to assume that the function is constant, $f(x) \approx c$
- This is pretty good because our area will become a rectangle, and we know how to calculate the area of a rectangle! $A = wh$

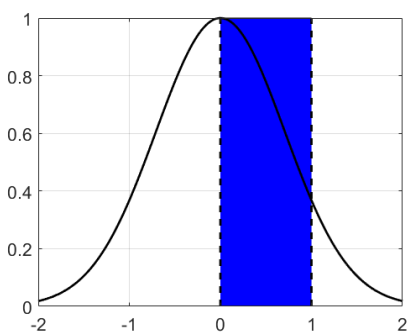
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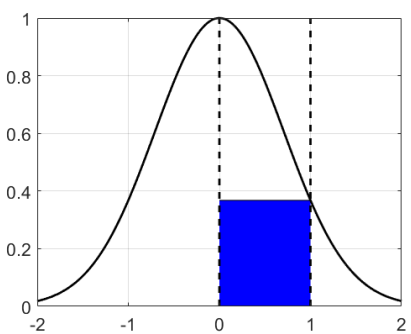
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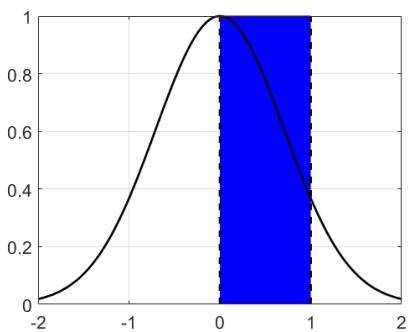
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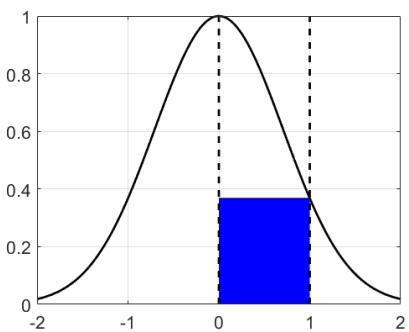


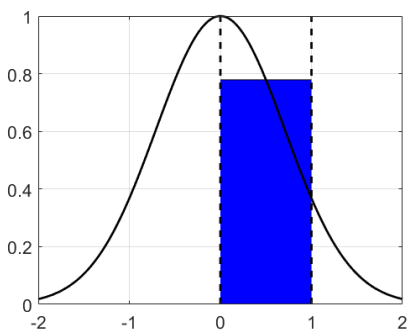


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  - **Right-hand rule**, $f(x) \approx f(b) \Rightarrow \int_a^b f(x)dx \approx (b-a)f(b)$
  - **Midpoint rule**, $f(x) \approx f(\frac{a+b}{2}) \Rightarrow \int_a^b f(x)dx \approx (b-a)f(\frac{a+b}{2})$







Trapezoidal Rule

- If we want to allow out function to actually change with varying x values, the next easiest thing we can do is assume the function is a line between $(a, f(a))$ and $(b, f(b))$

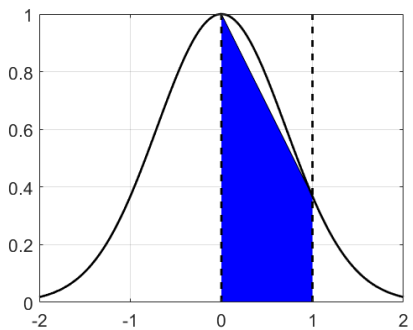
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- We could write this out as a function, but we can think of this as forming a trapezoid with our area. We know how to calculate the area of a trapezoid! $A = w\frac{h_1+h_2}{2}$
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Ex: $$\int_0^1 e^{-x^2}dx \approx (1-0)\frac{e^{-(0^2)}+e^{-(1^2)}}{2}$$



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- Now that we have seen interpolation, we can go another step further and find a parabola that goes through our edges and our midpoint $m = \frac{a+b}{2}$. We then find a surprisingly nice formula comes out of that!

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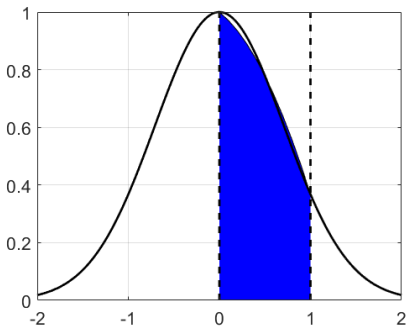
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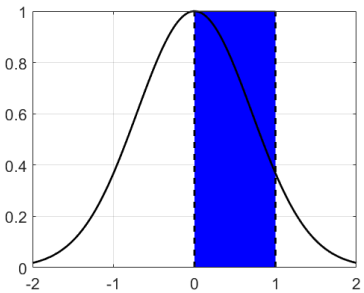
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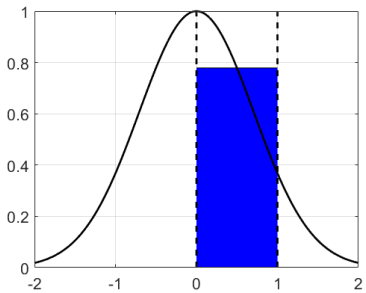
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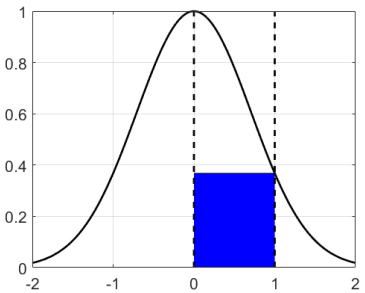


Comparisons

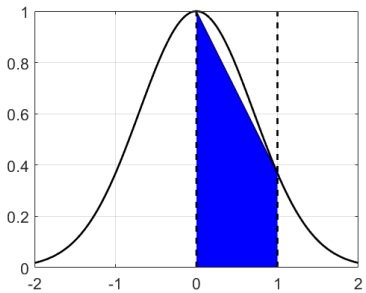
- Rectanglular Methods



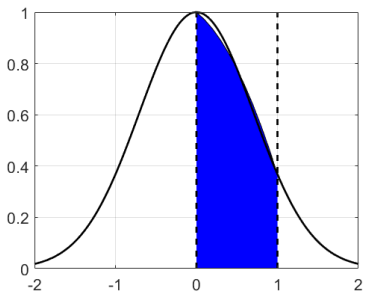




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- Simpson's Method



Riemann Sums

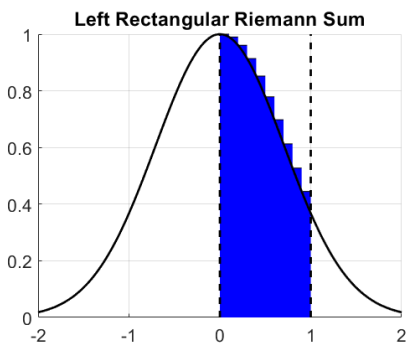
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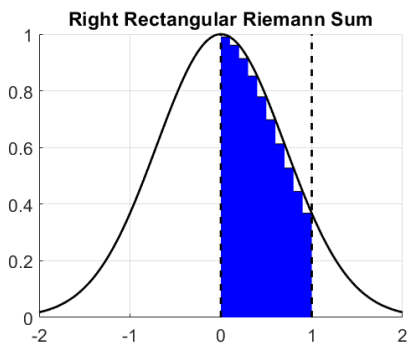
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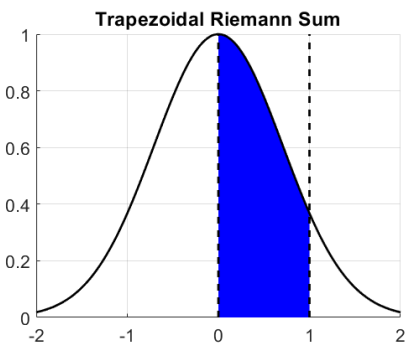
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- Here are our Riemann sums using 10 equal sized "slices"







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- This sequence can allow us to see an estimate of the antiderivative (plus or minus some constant C)

